

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath-six-vertex-app.com/. Learn this card is the memory jog.

NOTATION

R = Right = turn that face clockwise, looking at from outside the cube.
r = counter-clockwise. **2** = half turn. Lowercase **r f** = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run on its own unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's white color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'L'U



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U' R'U' y L'L'U



goes LEFT
U' L'L'U y' R'U'U'

Edge stuck in the wrong slot? Run either one to kick it out, then repeat it.

FIRST SOLVE

New here? Do the walkthrough once at cubepath-3x3.vercel.app/!
 Find a cube in the memory jog.
NOTATION
 R = right turn = turn that face clockwise, looking at from outside the cube.
 = counter-clockwise. 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube, left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns + trigger = a short algorithm you hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each where its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the empty right slot. B brings in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'L'U



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: Turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT
U' R'U' y L'U'L'U



goes LEFT
U' L'U'L'U y' R'U'R'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'



Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sun $\S$ 1 yellow, rest clockwise
RUR'U RU2R'



Anti-Sun $\S$ 1 yellow, rest counter-clockwise
RU2R' U'RU'R'



P4 $\S$ 2 yellow, headlights left
fRUR'U'f' FRUR'U'f'

OLL CORNERS - continued



Headlights $\S$ 2 yellow, headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\S$ 2 yellow, headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\S$ 2 yellow next to each other
rUR'U'f' FRUF'



Bowtie $\S$ 2 yellow diagonal
F'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, three is the worst.

Top not all yellow and nothing matches? Run Sun and read again.

TWO-LOOK

OLL CROSS - yellow edges

 **Line bar left-to-right**
fRUR'U'f'

 **Hook hook pointing front-right**
fRUR'U'f'

OLL CORNERS - yellow face

 **Sune** $\frac{1}{2}$ yellow; rest clockwise
RUR'U' RU2R'

 **Anti-Sune** $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' URUR'

 **Pi** $\frac{1}{2}$ yellow; headlights left
fRUR'U'f' FRUR'U'f'

OLL CORNERS

Headlights $\frac{3}{4}$ yellow; headlights at the back
R2DR'U2 RD'R'U2R'

 **2x Headlights** $\frac{3}{4}$ yellow; headlights both sides
RUR'U' RUR'R' URU2R'

 **Chameleon** $\frac{3}{4}$ yellow next to each other
rUR'U' FRF'

 **Bowtie** $\frac{3}{4}$ yellow diagonal
F'RUR'U'r' FR

FAILBACK
The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

ONE-LOOK PU ADJACENT CORNER SWAP exactly one face shows	ADJACENT CORNER SWAP continued
 • T1: headlights; pairs B-left F-left R'U'R' R'F R2UR'U' RUR'F'	 Ra 1. headlights; pairs F-left R'U'R' RUR DR'UR'D' R'U2R'
 Ja L solid; pairs B-right F-left R-front x R2FRF' RU2 r'U'r U2	 Rb 1. headlights; pairs B-left R2R RURU'R' F'R U2R'U2R
 Jb L solid; pairs B-left F-back RUR'F' RUR'U' R'F R2UR'R'	 Ga 1. headlights; pairs F-right R2UR'UR'U' RU'R2 U'D R'UR'D'
 Aa 1. headlights; pairs F-right R-front x L2D2 L'U'L D2 L'U'L'	 Gb 1. headlights; pairs R-back R'UR' UD' R2UR'UR'U' RU'R2D
 Ab 1. headlights; pairs B-right R-back x' L2D2 LUL' D2 LU'L	 Gc 1. headlights; pairs B-right R2UR'UR'U' RU'R2 UD' RU'R'D
 F1 L solid R'U'F' RUR'U' R'F R2UR'U'	 Gd 1. headlights; pairs R-front R'UR' UD' R2UR'UR' UR'UR2D'
One face shows headlight; two corners of that face match. Hold as drawn, run, then turn the top to finish.	Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PUZZLE ADJACENT CORNER SWAP ¹ exactly one face shows headlight.	ADJACENT CORNER SWAP ² continued
 T 1. headlights; pairs R-left, F-left R'U'L' R'F R2U'R'U' R'U'R'F'	 Ra 1. headlights; pairs D-front, R-left R'U'R'U' R'U'R D'R'U'R'D' R'U2R'
 Ja 1. solid; pairs B-right, F-left, R-front x R2FR'F' RU2 r'U'r U2	 Rb 1. headlights; pairs B-left, F-right R2F'RU'R'U' F'R U2R'U2R
 Ib 1. solid; pairs B-left, F-right, R-back R'U'R'F' RU'R'U' R'F R2U'R'	 Ga 1. headlights; pairs F-right, R-back R2UR'U'R'U' RU'R2 U'D' R'U'R'D'
 Aa 1. headlights; pairs F-right, B-front x L2D2 L'U'L D2 L'U'L'	 Gb 1. headlights; pairs R-back, R-front R'U'R'U'D' R2U'R'U'R'U' RU'R2D
 Ab 1. headlights; pairs B-right, R-back x' L2D2 LUL' D2 LU'L	 Gc 1. headlights; pairs B-right, R-front R2U'R'U'R'U' RU'R2 U'D' RU'R'D'
 F 1. solid R'U'F' RU'R'U' R'F R2U'R'U'U'	 Gd 1. headlights; pairs R-front, R-back R'U'R' U'D' R2U'R'U'R'U' UR'UR2D'

One face, two corners of two corners of that face match. Hold as described, run, then turns the top to finish.

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day — one is the mirror of the other.

[illegible]

ANALYZING BIG CUBES AND THE MAP

2 BIG CUBES 4x5 and 3x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. 3w = 2 big cubes wide. 3w = 3 layers. 2w = 2nd LAYER ONLY – never last 2 edges

4x4 PLL parity

$$\begin{matrix} R'w & U & L' & U & R' & F & R' & F' & R'w \\ 2R2 & U2 & 2R2 & U2 & 2R2 & U2 & 2R2 & U2 \end{matrix}$$
 slice out, run it, slice back – both cubes
2 edge pairs swapped – half the time

4x4 OLL parity = 1 edge pair flipped – half the time

$$\begin{matrix} R'w & U2 & x & R'w & U2 & R'w & U2 & R'w \\ R'w & U2 & x & R'w & U2 & R'w & U2 & R'w \end{matrix}$$
 slice out, run it, slice back – both cubes
one corner differs, 5x2 has no PLL parity

$$\begin{matrix} R'w & U2 & x & R'w & U2 & R'w & U2 & R'w \\ R'w & U2 & x & R'w & U2 & R'w & U2 & R'w \end{matrix}$$
 slice out, run it, slice back – both cubes
one corner differs, 5x2 has no PLL parity

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

Each letter turns a face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise = counter-clockwise, Z = half turn.

M – middle slice, turning the way L turns. x, y, z = the whole cube, turning the way R U L and F turn. Lowercase f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT

use it until you know None
Twist a yellow corner four turns at a time, turning only the top face between corners.

yellow faces RIGHT
$$\begin{matrix} D'R'D \\ x2 \end{matrix}$$

yellow faces FRONT
$$\begin{matrix} D'R'DR \\ x2 \end{matrix}$$

WORDS

algorithm = a fixed sequence of turns - trigger a short algorithm you hands run as one unit - sexy move = the R U R' U' - trigger - sledgehammer = the R' F R' F' turns - parity = a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5

Print at 100% / Actual size – do NOT fit to page. Two-sided, flip on the SHORT edge. The number of copies must be exactly 80 mm: _____

[illegible]